Evidence-selection counts

Bird noise systematic map

Working document · 27 August 2026

Scope: Counts below use records as the unit of reporting.

Records identified by database or search source

Database or search source	Records
Google Scholar — multilingual searches (nine exports × 100)	900
Scopus	635
Web of Science	478
OpenAlex	33
BASE (three exports: 9 + 5 + 2)	16
Total records identified	**2,062**

Check: 900 + 635 + 478 + 33 + 16 = 2,062.

Deduplication, screening, and final inclusion

Selection stage	Records	Reconciliation note
Records identified	2,062	Sum across all database and search-source exports
Duplicates removed with R	449	Removed before upload to Rayyan
Duplicates removed in Rayyan	3	Removed in Rayyan after R deduplication
Total duplicates removed	452	449 + 3
Unique records screened	1,610	2,062 − 452
Title-and-abstract exclusions	1,356	Reasons are not reported at this stage
Full texts sought	254	1,610 − 1,356
Full texts unavailable	3	Not included among assessed full texts
Full texts assessed	251	254 − 3
Full texts excluded	170	Reasons shown in the next section
Eligible records	81	251 − 170
Thesis records represented by included publications	4	Not counted as separate final records

Selection stage	Records	Reconciliation note
Final systematic-map records	**77**	**81 − 4**

Full-text exclusion reasons

Primary exclusion reason	Records
Wrong exposure	97
Wrong study design	46
Wrong publication type	12
Wrong comparator	9
Wrong language	3
Wrong outcome	2
Wrong population	1
Total full-text exclusions	**170**

Full text unavailable (n = 3) is a retrieval outcome and is therefore excluded from the 170 assessed full-text exclusions. For records with multiple Rayyan labels, Wrong study design received priority.

Arithmetic reconciliation

1. 2,062 − 452 = 1,610 unique records screened
2. 1,610 − 1,356 = 254 full texts sought
3. 254 − 3 = 251 full texts assessed
4. 251 − 170 = 81 eligible records
5. 81 − 4 = 77 final systematic-map records

Source files

Pre-deduplication exports: bird_map_decisions_08182026/before_deduplication_birds_records_040426/

Complete title-and-abstract screening decisions: Rayyan export

Full-text eligibility decisions: Rayyan export

Final processed dataset: Data/bird_clean_v2_utf8.csv

Appendix A. Records excluded after full-text assessment (n = 170)

Each record has one primary exclusion reason. Wrong study design received priority when a record had multiple active Rayyan labels. The three unavailable full texts were not assessed and are listed separately in Appendix B.

Rayyan ID	Reference	Primary reason
544368889	Jiménez-Vargas, GM (2019). Informe final para optar al título de Bióloga.	Wrong population
544326624	Akçay, C and Beecher, MD (2019). Multi-modal communication: song sparrows increase signal redundancy in noise. BIOLOGY LETTERS. https://doi.org/10.1098/rsbl.2019.0513	Wrong exposure
544326930	Antze, B and Koper, N (2018). Noisy anthropogenic infrastructure interferes with alarm responses in Savannah sparrows (Passerculus sandwichensis). ROYAL SOCIETY OPEN SCIENCE. https://doi.org/10.1098/rsos.172168	Wrong exposure
544326539	Arroyo-Solís, A and Castillo, JM and Figueroa, E and ... (2013). Experimental evidence for an impact of anthropogenic noise on dawn chorus timing in urban birds. Journal of Avian https://doi.org/10.1111/j.1600-048X.2012.05796.x	Wrong exposure
544326894	Bermúdez-Cuamatzin, E and López-Hernández, M and Campbell, J and Zuria, I and Slabbekoorn, H (2018). The role of singing style in song adjustments to fluctuating sound conditions: A comparative study on Mexican birds. BEHAVIOURAL PROCESSES. https://doi.org/10.1016/j.beproc.2018.04.005	Wrong exposure
544326456	Bermúdez-Cuamatzin, E and Riós-Chelén, AA and Gil, D and Garcia, CM (2012). Experimental evidence for real-time song frequency shift in response to urban noise in a passerine bird (vol 7, pg 36, 2010). BIOLOGY LETTERS. https://doi.org/10.1098/rsbl.2011.1225	Wrong exposure
544326563	Bermúdez-Cuamatzin, E and Ríos-Chelén, AA and ... (2011). Experimental evidence for real-time song frequency shift in response to urban noise in a passerine bird. Biology	Wrong exposure
544326583	Bermúdez-Cuamatzin, E and Slabbekoorn, H and Garcia, CM (2023). Spectral and temporal call flexibility of House Finches (Haemorhous mexicanus) from urban areas during experimental noise exposure. IBIS. https://doi.org/10.1111/ibi.13161	Wrong exposure
544326896	Bermúdez-Cuamatzin, E. and Ríos-Chelén, A.A. and Gil, D. and Garcia, C.M. (2012). Erratum: Experimental evidence for real-time song frequency shift in response to urban noise in a passerine bird (Biology Letters (2011)7 (36-38) DOI: 10.1098/rsbl.2010.0437). Biology Letters. https://doi.org/10.1098/rsbl.2011.1225	Wrong exposure
544326623	Blackburn, G and Ashton, BJ and Ridley, AR (2024). Evidence that multiple anthropogenic stressors cumulatively affect foraging and vigilance in an urban-living bird. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2024.02.014	Wrong exposure
544326403	Blackburn, G and Ashton, BJ and Thornton, A and Woodiss-Field, S and Ridley, AR (2023). Cognition mediates response to anthropogenic noise in wild Western Australian magpies (Gmynorhina tibicen dorsalis). GLOBAL CHANGE BIOLOGY. https://doi.org/10.1111/gcb.16975	Wrong exposure
544326582	Blumstein, DT and Whitaker, J and Kennen, J and Bryant, GA (2017). Do birds differentiate between white noise and deterministic chaos?. Ethology. https://doi.org/10.1111/eth.12702	Wrong exposure
544326357	Brumm, H and Todt, D (2002). Noise-dependent song amplitude regulation in a territorial songbird. Animal behaviour.	Wrong exposure
544326666	Budka, M (2025). Thrush nightingales adjust the peak frequency and structure of their songs in response to different types of experimental noise. SCIENTIFIC REPORTS. https://doi.org/10.1038/s41598-025-85991-3	Wrong exposure
544327019	Campo, JL and Gil, MG and Davila, SG (2005). Effects of specific noise and music stimuli on stress and fear levels of laying hens of several breeds. Applied Animal Behaviour Science.	Wrong exposure
544326430	Chou, TL and Krishna, A and Fossesca, M and Desai, A and Goldberg, J and Jones, S and Stephens, M and Basile, BM and Gall, MD (2023). Interspecific differences in the effects of masking and distraction on anti-predator behavior in suburban anthropogenic noise. PLOS ONE. https://doi.org/10.1371/journal.pone.0290330	Wrong exposure

Rayyan ID	Reference	Primary reason
544327139	Corbani, TL and Martin, JE and Healy, SD (2021). The impact of acute loud noise on the behavior of laboratory birds. Frontiers in Veterinary Science. https://doi.org/10.3389/fvets.2020.607632	Wrong exposure
544327085	Counter, S.A. (1985). Brain-stem evoked potentials and noise effects in seagulls. Comparative Biochemistry and Physiology -- Part A: Physiology. https://doi.org/10.1016/0300-9629(85)90916-8	Wrong exposure
544326787	Courter, JR and Perruci, RJ and McGinnis, KJ and Rainieri, JK (2020). Black-capped chickadees (Poecile atricapillus) alter alarm call duration and peak frequency in response to traffic noise. PLOS ONE. https://doi.org/10.1371/journal.pone.0241035	Wrong exposure
544326738	Damsky, J and Gall, MD (2017). Anthropogenic noise reduces approach of Black-capped Chickadee (Poecile atricapillus) and Tufted Titmouse (Baeolophus bicolor) to Tufted Titmouse mobbing calls. CONDOR. https://doi.org/10.1650/CONDOR-16-146.1	Wrong exposure
544326376	Davies, S. and Haddad, N. and Ouyang, J.Q. (2017). Stressful city sounds: Glucocorticoid responses to experimental traffic noise are environmentally dependent. Biology Letters. https://doi.org/10.1098/rsbl.2017.0276	Wrong exposure
544326377	des Aunay, GH and Slabbekoorn, H and Nagle, L and Passas, F and Nicolas, P and Draganoiu, TI (2014). Urban noise undermines female sexual preferences for low-frequency songs in domestic canaries. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2013.10.010	Wrong exposure
544326667	Dorado-Correa, AM and Zollinger, SA and ... (2018). Vocal plasticity in mallards: multiple signal changes in noise and the evolution of the Lombard effect in birds. Journal of Avian https://doi.org/10.1111/jav.01564	Wrong exposure
544327026	Duarte, R.H.L. and de Oliveira Passos, M.F. and Beirão, M.V. and Midamegbe, A. and Young, R.J. and de Azevedo, C.S. (2023). Noise interfere on feeding behaviour but not on food preference of saffron finches (Sicalis flaveola). Behavioural Processes. https://doi.org/10.1016/j.beproc.2023.104844	Wrong exposure
544326619	Efrati, A and Gutfreund, Y (2011). Early life exposure to noise alters the representation of auditory localization cues in the auditory space map of the barn owl. JOURNAL OF NEUROPHYSIOLOGY. https://doi.org/10.1152/jn.00078.2011	Wrong exposure
544326743	Evans, J.C. and Dall, S.R.X. and Kight, C.R. (2018). Effects of ambient noise on zebra finch vigilance and foraging efficiency. PLoS ONE. https://doi.org/10.1371/journal.pone.0209471	Wrong exposure
544326721	Flores, R and Penna, M and Wingfield, JC and Cuevas, E and Vásquez, RA and Quirici, V (2019). Effects of traffic noise exposure on corticosterone, glutathione and tonic immobility in chicks of a precocial bird. CONSERVATION PHYSIOLOGY. https://doi.org/10.1093/conphys/coz061	Wrong exposure
544326455	Foppen, K. and Pinxten, R. and Meijdam, M. and Eens, M. (2024). Artificial Light at Night Advances the Onset of Vocal Activity in Both Male and Female Great Tits During the Breeding Season, While Noise Pollution Has Less Impact and Only in Females. Animals. https://doi.org/10.3390/ani14223199	Wrong exposure
544327077	Gentry, KE and Derryberry, EP and Danner, RM and Danner, JE and Luther, DA (2017). Immediate signaling flexibility in response to experimental noise in urban, but not rural, white-crowned sparrows. ECOSPHERE. https://doi.org/10.1002/ecs2.1916	Wrong exposure
544326980	Grunst, ML and Grunst, AS and Pinxten, R and Eens, M (2021). Little parental response to anthropogenic noise in an urban songbird, but evidence for individual differences in sensitivity. SCIENCE OF THE TOTAL ENVIRONMENT. https://doi.org/10.1016/j.scitotenv.2020.144554	Wrong exposure
544326413	Grunst, ML and Grunst, AS and Pinxten, R and Eens, M (2021). Variable and consistent traffic noise negatively affect the sleep behavior of a free-living songbird. SCIENCE OF THE TOTAL ENVIRONMENT. https://doi.org/10.1016/j.scitotenv.2021.146338	Wrong exposure
544326501	Halfwerk, W and Bot, S and Buikx, J and van der Velde, M and Komdeur, J and ten Cate, C and Slabbekoorn, H (2011). Low-frequency songs lose their potency in noisy urban conditions. PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA. https://doi.org/10.1073/pnas.1109091108	Wrong exposure
544326462	Halfwerk, W and Holleman, LJM and ... (2011). Negative impact of traffic noise on avian reproductive success. Journal of applied https://doi.org/10.1111/j.1365-2664.2010.01914.x	Wrong exposure
544327086	Halfwerk, W and Slabbekoorn, H (2009). A behavioural mechanism explaining noise-dependent frequency use in urban birdsong. Animal behaviour.	Wrong exposure

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544326845	Hanna, D and Blouin-Demers, G and Wilson, DR and Mennill, DJ (2011). Anthropogenic noise affects song structure in red-winged blackbirds (Agelaius phoeniceus). JOURNAL OF EXPERIMENTAL BIOLOGY. https://doi.org/10.1242/jeb.060194	Wrong exposure
544326725	Hennigar, B and Ethier, JP and Wilson, DR (2019). Experimental traffic noise attracts birds during the breeding season. Behavioral Ecology.	Wrong exposure
544326779	Hubbard, L. and King, W. and Vu, A. and Blumstein, D.T. (2015). Heterospecific nonalarm vocalizations enhance risk assessment in common mynas. Behavioral Ecology. https://doi.org/10.1093/beheco/arv002	Wrong exposure
544326458	Iasiello, L and Colombelli-Négrel, D (2023). Noisy neighbours: effects of construction noises on nesting seabirds. MARINE AND FRESHWATER RESEARCH. https://doi.org/10.1071/MF22138	Wrong exposure
544368809	II, C Sayers and Moreland, C and Morgan, H and Arévalo, JE (2019). Efecto de corto plazo del ruido por tráfico sobre coros de aves en un bosque nuboso neotropical.. Zeledonia.	Wrong exposure
544368831	Izarra, FA Riera (n.d.). Tasa de consumo de oxígeno de Zonotrichia capensis expuesto a distintos playback. saber.ucv.ve.	Wrong exposure
544326910	Jones, R.B. (1986). Background auditory stimulation and tonic immobility in the domestic fowl. IRCS Medical Science.	Wrong exposure
544326461	Kern, JM and Radford, AN (2016). Anthropogenic noise disrupts use of vocal information about predation risk. ENVIRONMENTAL POLLUTION. https://doi.org/10.1016/j.envpol.2016.08.049	Wrong exposure
544326514	LaZerte, SE and Slabbekoorn, H and Otter, KA (2016). Learning to cope: vocal adjustment to urban noise is correlated with prior experience in black-capped chickadees. PROCEEDINGS OF THE ROYAL SOCIETY B-BIOLOGICAL SCIENCES. https://doi.org/10.1098/rspb.2016.1058	Wrong exposure
544326385	LaZerte, SE and Slabbekoorn, H and Otter, KA (2017). Territorial black-capped chickadee males respond faster to high- than to low-frequency songs in experimentally elevated noise conditions. PEERJ. https://doi.org/10.7717/peerj.3257	Wrong exposure
544326592	Lehnardt, Y. and Klein, T. and Barber, J.R. and Berger-Tal, O. (2023). Experimentally Broadcasted Wind-turbine Sound Drastically Alters Songbirds' Habitat Selection and Vocal Communication in a Natural Environment. Proceedings of Forum Acusticum.	Wrong exposure
544326485	Leonard, ML and Horn, AG (2012). Ambient noise increases missed detections in nestling birds. Biology letters.	Wrong exposure
544326826	Liu, QX and Slabbekoorn, H and Riebel, K (2020). Zebra finches show spatial avoidance of near but not far distance traffic noise. BEHAVIOUR. https://doi.org/10.1163/1568539X-bja10004	Wrong exposure
544326662	Lucass, C and Eens, M and Müller, W (2016). When ambient noise impairs parent-offspring communication. ENVIRONMENTAL POLLUTION. https://doi.org/10.1016/j.envpol.2016.03.015	Wrong exposure
544326682	Manabe, K and Sadr, EI and Dooling, RJ (1998). Control of vocal intensity in budgerigars (Melopsittacus undulatus): Differential reinforcement of vocal intensity and the Lombard effect. The Journal of the Acoustical Society ….	Wrong exposure
544326960	Matyjasiak, P and Chacinska, P and Ksiazka, P (2023). Anthropogenic noise interacts with the predation risk assessment in a free-ranging bird. CURRENT ZOOLOGY. https://doi.org/10.1093/cz/zoad019	Wrong exposure
544326849	Matyjasiak, P and Ksiazka, P and Chacinska, P (2026). Beyond noise playback: the effect of an audible and visible sound source on bird flushing behavior. JOURNAL OF ETHOLOGY. https://doi.org/10.1007/s10164-025-00864-6	Wrong exposure
544326364	McLaughlin, KE and Kunc, HP (2013). Experimentally increased noise levels change spatial and singing behaviour. Biology letters.	Wrong exposure
544327029	McMullen, H and Schmidt, R and Kunc, HP (2014). Anthropogenic noise affects vocal interactions. BEHAVIOURAL PROCESSES. https://doi.org/10.1016/j.beproc.2013.12.001	Wrong exposure
544368801	Mendoza, NE Chavez (2020). Ruido urbano y comunicación acústica en aves: evaluando el efecto del ruido en la conducta territorial de un ave suboscina.	Wrong exposure
544326404	Merrall, ES and Evans, KL (2020). Anthropogenic noise reduces avian feeding efficiency and increases vigilance along an urban-rural gradient regardless of species' tolerances to urbanisation. JOURNAL OF AVIAN BIOLOGY. https://doi.org/10.1111/jav.02341	Wrong exposure
544327118	Monserrat, S and Constantino, MG and Bibiana, M (2026). Simulated fireworks disrupt behaviour, but apparently not glucose or corticosterone levels in captive Atlantic canaries Serinus canaria. IBIS. https://doi.org/10.1111/ibi.70065	Wrong exposure
544326684	Montague, MJ and Danek-Gontard, M and Kunc, HP (2013). Phenotypic plasticity affects the response of a sexually selected trait to anthropogenic noise. BEHAVIORAL ECOLOGY. https://doi.org/10.1093/beheco/ars169	Wrong exposure

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544326948	Moore, J.L. and Roach, S. (2026). Anthropogenic noise impacts territorial defense in a non-urban songbird. Behavioural Processes. https://doi.org/10.1016/j.beproc.2026.105382	Wrong exposure
544326443	Moriya, M and Senzaki, M and Kitazawa, M and Kawamura, K and … (2026). Increased vulnerability of alert responses to combined call sequences under anthropogenic noise in bird communication. … Science and Pollution …. https://doi.org/10.1007/s11356-026-37457-w	Wrong exposure
544326654	Muñoz-Santos, I and Ríos-Chelén, AA (2023). Vermilion flycatchers avoid singing during sudden peaks of anthropogenic noise. ACTA ETHOLOGICA. https://doi.org/10.1007/s10211-022-00409-x	Wrong exposure
544326685	Muñoz-Santos, I and Sosa-López, JR and Rios-Chelen, AA (2025). Non-masking noise suppresses bird song. BEHAVIOURAL PROCESSES. https://doi.org/10.1016/j.beproc.2025.105239	Wrong exposure
544326551	Naguib, M and van Oers, K and Braakhuis, A and Griffioen, M and de Goede, P and Waas, JR (2013). Noise annoys: effects of noise on breeding great tits depend on personality but not on noise characteristics. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2013.02.015	Wrong exposure
544326699	Okanoya, K and Dooling, RJ (1987). Hearing in passerine and psittacine birds: a comparative study of absolute and masked auditory thresholds. Journal of Comparative Psychology.	Wrong exposure
544326914	Orlando, G and Passarotto, A and Morosinotto, C and Dominoni, DM and Karell, P (2025). Experimental Exposure to Noise Affects Hunting Behavior Already From a Young Age in a Nocturnal Acoustic Predator. ECOLOGY AND EVOLUTION. https://doi.org/10.1002/ece3.72171	Wrong exposure
544326715	Osbrink, A and Meatte, MA and Tran, A and Herranen, KK and Meek, L and Murakami-Smith, M and Ito, J and Bhadra, S and Nunnenkamp, C and Templeton, CN (2021). Traffic noise inhibits cognitive performance in a songbird. PROCEEDINGS OF THE ROYAL SOCIETY B-BIOLOGICAL SCIENCES. https://doi.org/10.1098/rspb.2020.2851	Wrong exposure
544326431	Ozkan, K and Langley, JM and Talbott, JW and Kleist, NJ and Francis, CD (2024). Divergent effects of short-term and continuous anthropogenic noise exposure on Western Bluebird parental care behavior. PEERJ. https://doi.org/10.7717/peerj.18558	Wrong exposure
544326776	Passos, MFD and Beirao, MV and Midamegbe, A and Duarte, RHL and Young, RJ and de Azevedo, CS (2020). Impacts of noise pollution on the agonistic interactions of the saffron finch (Sicalis flaveola Linnaeus, 1766). BEHAVIOURAL PROCESSES. https://doi.org/10.1016/j.beproc.2020.104222	Wrong exposure
544326457	Pettinga, D and Kennedy, J and Proppe, DS (2016). Common urban birds continue to perceive predator calls that are overlapped by road noise. URBAN ECOSYSTEMS. https://doi.org/10.1007/s11252-015-0498-9	Wrong exposure
544326964	Potvin, DA and Mulder, RA (2013). Immediate, independent adjustment of call pitch and amplitude in response to varying background noise by silvereyes (Zosterops lateralis). BEHAVIORAL ECOLOGY. https://doi.org/10.1093/beheco/art075	Wrong exposure
544326409	Ratnayake, CP and Zhou, Y and Pell, FSED and Potvin, DA and Radford, AN and Magrath, RD (2021). Visual obstruction, but not moderate traffic noise, increases reliance on heterospecific alarm calls. BEHAVIORAL ECOLOGY. https://doi.org/10.1093/beheco/arab051	Wrong exposure
544326663	Reichard, DG and Atwell, JW and Pandit, MM and Cardoso, GC and Price, TD and Ketterson, ED (2020). Urban birdsongs: higher minimum song frequency of an urban colonist persists in a common garden experiment. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2020.10.007	Wrong exposure
544368805	Resende, DA de Lara and … (2022). Efeito do ruído urbano na comunicação acústica da choca-de-asa-vermelha (Thamnophilus Torquatus): uma manipulação experimental. Programa de ….	Wrong exposure
544326511	Ryals, BM and Dooling, RJ and Westbrook, E and Dent, ML and … (1999). Avian species differences in susceptibility to noise exposure. Hearing research.	Wrong exposure
544326477	Ríos-Chelén, AA and Cuatianquiz-Lima, C and Bautista, A and Martínez-Gómez, M (2018). No reliable evidence for immediate noise-induced song flexibility in a suboscine. URBAN ECOSYSTEMS. https://doi.org/10.1007/s11252-017-0690-1	Wrong exposure
544326968	Ríos-Chelén, AA and Lee, GC and Patricelli, GL (2016). A comparison between two ways to measure minimum frequency and an experimental test of vocal plasticity in red-winged blackbirds in response to noise. BEHAVIOUR. https://doi.org/10.1163/1568539X-00003390	Wrong exposure
544326665	Slabbekoorn, H and Ellers, J and Smith, TB (2002). Birdsong and sound transmission: the benefits of reverberations. The Condor. https://doi.org/10.1650/0010-5422(2002)104[0564:BASTTB	Wrong exposure

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544326754	Smith, S and Evans, B and Luther, DA (2025). An experiment of how error rates shape animal communication and the influence of anthropogenic noise. ROYAL SOCIETY OPEN SCIENCE. https://doi.org/10.1098/rsos.251937	Wrong exposure
544326378	Sokołowska, E and Staniewicz, A and Muszyńska, A and Budka, M (2026). Temporal avoidance of interfering noise by common chiffchaff. BEHAVIORAL ECOLOGY AND SOCIOBIOLOGY. https://doi.org/10.1007/s00265-026-03704-w	Wrong exposure
544326736	Swaddle, JP and Page, LC (2007). High levels of environmental noise erode pair preferences in zebra finches: implications for noise pollution. Animal behaviour.	Wrong exposure
544326706	Sweet, KA and Sweet, BP and Gomes, DGE and Francis, CD and Barber, JR (2022). Natural and anthropogenic noise increase vigilance and decrease foraging behaviors in song sparrows. BEHAVIORAL ECOLOGY. https://doi.org/10.1093/beheco/arab141	Wrong exposure
544326802	Taylor, A. and Berryman, J.C. and Burgess, D.J.H. and Hill, D. and Mann, D. (1983). The effects of environmental change on chicks'sresponses to a familiar stimulus. QUARTERLY JOURNAL OF EXPERIMENTAL PSYCHOLOGY SECTION B-COMPARATIVE AND PHYSIOLOGICAL PSYCHOLOGY. https://doi.org/10.1080/14640748308400894	Wrong exposure
544326470	Templeton, CN and O'Connor, A and Strack, S and Meraz, F and Herranen, K (2023). Traffic noise inhibits inhibitory control in wild-caught songbirds. ISCIENCE. https://doi.org/10.1016/j.isci.2023.106650	Wrong exposure
544326525	Templeton, CN and Zollinger, SA and Brumm, H (2016). Traffic noise drowns out great tit alarm calls. Current Biology.	Wrong exposure
544326693	THES (1997). Heat shock protein expression in response to stress and diabetes.	Wrong exposure
544326447	Tilgar, V and Vligipuu, R (2024). Effects of noise on the nestling feeding behaviour of Great Tits do not depend on noise characteristics. ACTA ORNITHOLOGICA. https://doi.org/10.3161/00016454AO2024.59.1.006	Wrong exposure
544326441	Veon, JT and McClung, MR (2023). Disturbance of wintering waterbirds by simulated road traffic noise in Arkansas wetlands. JOURNAL OF WILDLIFE MANAGEMENT. https://doi.org/10.1002/jwmg.22387	Wrong exposure
544326783	Verzijden, MN and Ripmeester, EAP and ... (2010). Immediate spectral flexibility in singing chiffchaffs during experimental exposure to highway noise. Journal of	Wrong exposure
544326656	Vignal, C and Attia, J and Mathevon, N and Beauchaud, M (2004). Background noise does not modify song-induced genic activation in the bird brain. Behavioural brain research.	Wrong exposure
544326649	Waldinger, J and Warrington, MH and Ellison, K and Koper, N (2024). Song adjustments only partially restore effective communication among Baird's sparrows, Centronyx bairdii, exposed to oil well drilling noise. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2024.02.010	Wrong exposure
544326832	Winfield, MA and Murakami-Smith, M and Nunnenkamp, C and Templeton, CN (2026). Do Continuous and Intermittent Traffic Noise Have Similar Negative Impacts on Zebra Finch Cognitive Performance?. ETHOLOGY. https://doi.org/10.1111/eth.70078	Wrong exposure
544326902	Yang, XJ and Slabbekoorn, H (2014). Timing vocal behavior: Lack of temporal overlap avoidance to fluctuating noise levels in singing Eurasian wrens. BEHAVIOURAL PROCESSES. https://doi.org/10.1016/j.beproc.2014.10.002	Wrong exposure
544327044	Zhou, Y and Radford, AN and Magrath, RD (2019). Why does noise reduce response to alarm calls? Experimental assessment of masking, distraction and greater vigilance in wild birds. Functional Ecology. https://doi.org/10.1111/1365-2435.13333	Wrong exposure
544326767	Zhou, Y and Radford, AN and Magrath, RD (2024). Noise constrains heterospecific eavesdropping more than conspecific reception of alarm calls. BIOLOGY LETTERS. https://doi.org/10.1098/rsbl.2023.0410	Wrong exposure
544326668	Zhou, Y and Radford, AN and Magrath, RD (2025). The effect of temporal masking on alarm call communication in wild superb fairy-wrens. ANIMAL BEHAVIOUR. https://doi.org/10.1016/j.anbehav.2024.10.010	Wrong exposure
544326536	Zollinger, SA and Goller, F and Brumm, H (2011). Metabolic and Respiratory Costs of Increasing Song Amplitude in Zebra Finches. PLOS ONE. https://doi.org/10.1371/journal.pone.0023198	Wrong exposure
544326957	Zwart, MC and Dunn, JC and McGowan, PJK and ... (2016). Wind farm noise suppresses territorial defense behavior in a songbird. Behavioral	Wrong exposure
544326556	Önsal, Ç and Yelimlies, A and Akçay, Ç (2022). Aggression and multi-modal signaling in noise in a common urban songbird. BEHAVIORAL ECOLOGY AND SOCIOBIOLOGY. https://doi.org/10.1007/s00265-022-03207-4	Wrong exposure
544368751	Бусловская, ЛК and Ковтуненко, АЮ and ... (2010). Адаптация кур к факторам промышленного содержания. Региональные	Wrong exposure

Rayyan ID	Reference	Primary reason
544326990	Brown, AL (1990). Measuring the effect of aircraft noise on sea birds. Environment international.	Wrong comparator
544326611	Connelly, F and Johnsson, RD and Mulder, RA and Hall, ML and Lesku, JA (2024). Experimental playback of urban noise does not affect cognitive performance in captive Australian magpies. BIOLOGY OPEN. https://doi.org/10.1242/bio.060535	Wrong comparator
544326476	des Aunay, GH and Grenna, M and Slabbekoorn, H and Nicolas, P and Nagle, L and Leboucher, G and Malacarne, G and Draganoiu, TI (2017). Negative impact of urban noise on sexual receptivity and clutch size in female domestic canaries. ETHOLOGY. https://doi.org/10.1111/eth.12659	Wrong comparator
544326764	Goto, H and de Framond, L and Leitner, S and Brumm, H (2023). Bursts of white noise trigger song in domestic Canaries. JOURNAL OF ORNITHOLOGY. https://doi.org/10.1007/s10336-023-02070-y	Wrong comparator
544327001	Grubb, T.G. and Pater, L.L. and Gatto, A.E. and Delaney, D.K. (2013). Response of nesting northern goshawks to logging truck noise in northern Arizona. Journal of Wildlife Management. https://doi.org/10.1002/jwmg.607	Wrong comparator
544326958	Liu, QX and Gelok, E and Fontein, K and Slabbekoorn, H and Riebel, K (2022). An experimental test of chronic traffic noise exposure on parental behaviour and reproduction in zebra finches. BIOLOGY OPEN. https://doi.org/10.1242/bio.059183	Wrong comparator
544326856	McFarlane, J.M. and Curtis, S.E. and Shanks, R.D. and Carmer, S.G. (1989). MULTIPLE CONCURRENT STRESSORS IN CHICKS .1. EFFECT ON WEIGHT-GAIN, FEED-INTAKE, AND BEHAVIOR. POULTRY SCIENCE. https://doi.org/10.3382/ps.0680501	Wrong comparator
544326363	Moseley, DL and Derryberry, GE and Phillips, JN and Danner, JE and Danner, RM and Luther, DA and Derryberry, EP (2018). Acoustic adaptation to city noise through vocal learning by a songbird. PROCEEDINGS OF THE ROYAL SOCIETY B-BIOLOGICAL SCIENCES. https://doi.org/10.1098/rspb.2018.1356	Wrong comparator
544326772	Zollinger, SA and Slater, PJB and ... (2017). Higher songs of city birds may not be an individual response to noise. Proceedings of the	Wrong comparator
544327000	Rosa, P and Swider, CR and Leston, L and Koper, N (2015). Disentangling Effects of Noise from Presence of Anthropogenic Infrastructure: Design and Testing of System for Large-Scale Playback Experiments. WILDLIFE SOCIETY BULLETIN. https://doi.org/10.1002/wsb.546	Wrong outcome
544326482	Werrell, AK and Klug, PE and Lipcius, RN and Swaddle, JP (2021). A Sonic Net reduces damage to sunflower by blackbirds (Icteridae): Implications for broad-scale agriculture and crop establishment. CROP PROTECTION. https://doi.org/10.1016/j.cropro.2021.105579	Wrong outcome
596897772	Alissa R. Petrelli (2017). Influences of Anthropogenic Noise on Flight Initiation Distance, Foraging Behavior, And Feeder Community Structure of Wild Birds. https://doi.org/10.15368/theses.2018.90	Wrong study design
544327036	Blesdoe, EK and Blumstein, DT (2014). What is the sound of fear? Behavioral responses of white-crowned sparrows Zonotrichia leucophrys to synthesized nonlinear acoustic phenomena. CURRENT ZOOLOGY. https://doi.org/10.1093/czoolo/60.4.534	Wrong study design
596897768	Bronwen Hennigar (2018). Effects of anthropogenic noise and light on the vocal and spatial behaviour of birds.	Wrong study design
544327023	Brumm, H (2004). The impact of environmental noise on song amplitude in a territorial bird. Journal of animal ecology.	Wrong study design
544326479	Brumm, H and Goymann, W and Derégnaucourt, S and Geberzahn, N and Zollinger, SA (2021). Traffic noise disrupts vocal development and suppresses immune function. SCIENCE ADVANCES. https://doi.org/10.1126/sciadv.abe2405	Wrong study design
544368826	Campos, RSO (2020). Influência da urbanização no comportamento animal: há alterações na agressividade e na vocalização de uma ave tropical?.	Wrong study design
544368823	CUAMATZIN, EB (2014). CONSECUENCIAS DEL RUIDO AMBIENTAL EN LA COMUNICACIÓN ACÚSTICA DEL GORRIÓN MEXICANO (Carpodacus mexicanus).	Wrong study design
544326929	Curry, CM and Brisay, PG Des and Rosa, P and Koper, N (2018). Noise source and individual physiology mediate effectiveness of bird songs adjusted to anthropogenic noise. Scientific Reports.	Wrong study design
544327122	Dooling, R.J. and Saunders, J.C. (1974). Threshold shift produced by continuous noise exposure in the parakeet (Melopsittacus undulatus). Journal of the Acoustical Society of America. https://doi.org/10.1121/1.1919943	Wrong study design
544326545	Ellis, D.H. and Ellis, C.H. and Mindell, D.P. (1991). Raptor responses to low-level jet aircraft and sonic booms. Environmental Pollution. https://doi.org/10.1016/0269-7491(91)90026-S	Wrong study design

Rayyan ID	Reference	Primary reason
544326875	Francis, C.D. and Ortega, C.P. and Kennedy, R.I. and Nylander, P.J. (2012). Are nest predators absent from noisy areas or unable to locate nests?. Ornithological Monographs. https://doi.org/10.1525/om.2012.74.1.101	Wrong study design
544326574	Francis, CD and Ortega, CP and Cruz, A (2009). Noise pollution changes avian communities and species interactions. Current biology.	Wrong study design
544326690	Francis, CD and Ortega, CP and Cruz, A (2011). Different behavioural responses to anthropogenic noise by two closely related passerine birds. BIOLOGY LETTERS. https://doi.org/10.1098/rsbl.2011.0359	Wrong study design
596897773	Gabrielle Sá Melo Winandy (2020). Consequences of anthropogenic noise to song elaboration, signal value and reproductive success: field and captive studies in neotropical Bananaquits and Bengalese finches.	Wrong study design
544327018	Grabarczyk, E.E. and Araya-Salas, M. and Vonhof, M.J. and Gill, S.A. (2020). Anthropogenic noise affects female, not male house wren response to change in signaling network. Ethology. https://doi.org/10.1111/eth.13085	Wrong study design
544326825	Grabarczyk, EE and Gill, SA (2019). Anthropogenic noise affects male house wren response to but not detection of territorial intruders. PLOS ONE. https://doi.org/10.1371/journal.pone.0220576	Wrong study design
544326863	Grabarczyk, EE and Gill, SA (2020). Anthropogenic noise masking diminishes house wren (Troglodytes aedon) song transmission in urban natural areas. BIOACOUSTICS-THE INTERNATIONAL JOURNAL OF ANIMAL SOUND AND ITS RECORDING. https://doi.org/10.1080/09524622.2019.1621209	Wrong study design
544326652	Grace, MK and Anderson, RC (2015). No frequency shift in the "D" notes of Carolina chickadee calls in response to traffic noise. BEHAVIORAL ECOLOGY AND SOCIOBIOLOGY. https://doi.org/10.1007/s00265-014-1838-0	Wrong study design
544326625	Grade, AM and Sieving, KE (2016). When the birds go unheard: highway noise disrupts information transfer between bird species. Biology letters.	Wrong study design
544327132	Gross, K and Pasinelli, G and Kunc, HP (2010). Behavioral Plasticity Allows Short-Term Adjustment to a Novel Environment. AMERICAN NATURALIST. https://doi.org/10.1086/655428	Wrong study design
544326976	Grunst, A.S. and Grunst, M.L. and Raap, T. and Pinxten, R. and Eens, M. (2023). Anthropogenic noise and light pollution additively affect sleep behaviour in free-living birds in sex- and season-dependent fashions. Environmental Pollution. https://doi.org/10.1016/j.envpol.2022.120426	Wrong study design
544326614	Habib, L and Bayne, EM and Boutin, S (2007). Chronic industrial noise affects pairing success and age structure of ovenbirds Seiurus aurocapilla. JOURNAL OF APPLIED ECOLOGY. https://doi.org/10.1111/j.1365-2664.2006.01234.x	Wrong study design
544327131	Habib, Lucas David (2006). Effects of chronic industrial noise disturbance on boreal forest songbirds ... : Chronic industrial noise and boreal songbirds ...	Wrong study design
544326636	KLUMP, GM and LANGEMANN, U (1995). COMODULATION MASKING RELEASE IN A SONGBIRD. HEARING RESEARCH. https://doi.org/10.1016/0378-5955(95)00087-K	Wrong study design
544326916	Levenhagen, MJ and McClure, CJW and ... (2021). Does experimentally quieting traffic noise benefit people and birds?. Ecology and	Wrong study design
544326786	Liang, SH and Jen, CH and Lee, LL and Chen, CC and Shieh, BS (2024). Savanna Nightjars (Caprimulgus affinis stictomus) adjust calling height to gain amplitude advantage in urban environments. JOURNAL OF ORNITHOLOGY. https://doi.org/10.1007/s10336-023-02142-z	Wrong study design
544326602	Long, AM and Colón, MR and Bosman, JL and McFarland, TM and Locatelli, AJ and Stewart, LR and Mathewson, HA and Newnam, JC and Morrison, ML (2017). Effects of Road Construction Noise on Golden-Cheeked Warblers: An Update. WILDLIFE SOCIETY BULLETIN. https://doi.org/10.1002/wsb.777	Wrong study design
544326712	Luther, DA and Danner, R and Danner, J and Gentry, K and Derryberry, EP (2017). The relative response of songbirds to shifts in song amplitude and song minimum frequency. BEHAVIORAL ECOLOGY. https://doi.org/10.1093/beheco/arw172	Wrong study design
544327033	Luther, DA and Phillips, J and Derryberry, EP (2016). Not so sexy in the city: urban birds adjust songs to noise but compromise vocal performance. Behavioral Ecology.	Wrong study design
544369020	Manfred, G (2016). Rôle du chant dans la communication intra-et intersexuelle chez les oiseaux chanteurs: Effets du bruit urbain sur la réponse au chant des femelles de canari	Wrong study design
544368829	Miranda, C Hamilton-West (2005). Efectos del ruido en la conducta vocal de machos de Eupsophus emiliopugini (Amphibia, Leptodactylidae).	Wrong study design

Rayyan ID	Reference	Primary reason
544326542	Mockford, EJ and Marshall, RC (2009). Effects of urban noise on song and response behaviour in great tits. Proceedings of the Royal ….	Wrong study design
544327126	Montenegro, C and Service, WD and Scully, EN and Mischler, SK and Sahu, PK and Benowicz, TJ and Fox, KVR and Sturdy, CB (2021). The impact of anthropogenic noise on individual identification via female song in Black-capped chickadees (Poecile atricapillus). SCIENTIFIC REPORTS. https://doi.org/10.1038/s41598-021-96504-3	Wrong study design
544327021	Passarotto, A and Morosinotto, C and … (2025). Experimental noise and light pollution alter prey detection in a nocturnal bird of prey. Journal of Animal …. https://doi.org/10.1111/1365-2656.70062	Wrong study design
544326780	Raap, T and Pinxten, R and Casasole, G and Dehnhard, N and Eens, M (2017). Ambient anthropogenic noise but not light is associated with the ecophysiology of free-living songbird nestlings. SCIENTIFIC REPORTS. https://doi.org/10.1038/s41598-017-02940-5	Wrong study design
544326681	Ríos-Chelén, AA and McDonald, AN and Berger, A and … (2017). Do birds vocalize at higher pitch in noise, or is it a matter of measurement?. Behavioral Ecology and …. https://doi.org/10.1007/s00265-016-2243-7	Wrong study design
544327079	Sheldon, EL and Ironside, JE and de Vere, N and Marshall, RC (2020). Singing under glass: rapid effects of anthropogenic habitat modification on song and response behaviours in an isolated house sparrow Passer domesticus population. JOURNAL OF AVIAN BIOLOGY. https://doi.org/10.1111/jav.02248	Wrong study design
544326842	Slabbekoorn, H. and Peet, M. (2003). Birds sing at a higher pitch in urban noise. Nature. https://doi.org/10.1038/424267a	Wrong study design
596897776	Steffi LaZerte (2015). Sounds of the city: The effects of urbanization and noise on mountain and black-capped chickadee communication.. https://doi.org/10.24124/2015/bpgub1059	Wrong study design
544326359	Stone, E (2000). Separating the noise from the noise: a finding in support of the “niche hypothesis,” that birds are influenced by human-induced noise in natural habitats. Anthrozoös. https://doi.org/10.2752/089279300786999680	Wrong study design
544326564	Tempel, DJ and Gutiérrez, RJ (2003). Fecal corticosterone levels in California spotted owls exposed to low-intensity chainsaw sound. WILDLIFE SOCIETY BULLETIN.	Wrong study design
544327119	Van Donselaar, JL and Atma, JL and Kruyf, ZA and LaCroix, HN and Proppe, DS (2018). Urbanization alters fear behavior in black-capped chickadees. URBAN ECOSYSTEMS. https://doi.org/10.1007/s11252-018-0783-5	Wrong study design
544327064	Viigipuu, R and Mägi, M and Tilgar, V (2023). Great tits alter incubation behaviour in noisy environments. JOURNAL OF ETHOLOGY. https://doi.org/10.1007/s10164-022-00765-y	Wrong study design
544326766	Weaver, M and Hutton, P and McGraw, KJ (2019). Urban house finches (Haemorhous mexicanus) are less averse to novel noises, but not other novel environmental stimuli, than rural birds. BEHAVIOUR. https://doi.org/10.1163/1568539X-00003571	Wrong study design
544326922	Wheeler, Mariette (2009). The effects of human disturbance on the seabirds and seals at sub-Antarctic Marion Island.	Wrong study design
544368507	Japanese-language record (1988). Effects of noise exposure on avian auditory organs. Ear Research Japan.	Wrong study design
544327127	Brumm, H and Zollinger, SA (2013). Avian vocal production in noise. Animal communication and noise. https://doi.org/10.1007/978-3-642-41494-7_7	Wrong publication type
544327054	Dooling, R.J. and Brittan-Powell, B. (2006). The effects of noise on the hearing and vocalizations of birds. Institute of Noise Control Engineering of the USA - 35th International Congress and Exposition on Noise Control Engineering, INTER-NOISE 2006.	Wrong publication type
544327060	Dooling, RJ and Buehler, D and Leek, MR and Popper, AN (2019). The impact of urban and traffic noise on birds. Acoustics Today.	Wrong publication type
544368854	Gomes, MV da Costa (2024). Efeitos da poluição sonora na comunicação de espécies urbanas. Revista Tópicos.	Wrong publication type
544327050	Halfwerk, W and Lohr, B and Slabbekoorn, H (2018). Impact of Man-Made Sound on Birds and Their Songs. EFFECTS OF ANTHROPOGENIC NOISE ON ANIMALS. https://doi.org/10.1007/978-1-4939-8574-6_8	Wrong publication type
544326945	Injaian, AS and Taff, CC and Patricelli, GL (2018). Experimental Anthropogenic Noise Impacts Parental Behavior, and Nestling Growth and Oxidative Stress in a Non-urban Bird. INTEGRATIVE AND COMPARATIVE BIOLOGY.	Wrong publication type
544326947	Templeton, CN (2025). Noisy nests: Early-life noise exposure impacts songbird fitness. LEARNING & BEHAVIOR. https://doi.org/10.3758/s13420-024-00654-z	Wrong publication type

Rayyan ID	Reference	Primary reason
544326719	Xu, X. and Feng, J. and Jeon J.Y. (2020). Mapping the tourism transportation noise exposure to natural quiet and predicting potential effects on the threatened black-necked cranes in dashanbao protected areas. Proceedings of 2020 International Congress on Noise Control Engineering, INTER-NOISE 2020.	Wrong publication type
544326554	Zollinger, SA and Brumm, H (2018). Effects of Experimental Traffic Noise Exposure on Avian Health and Fitness. INTEGRATIVE AND COMPARATIVE BIOLOGY.	Wrong publication type
544368504	Japanese-language record (2024). Avian sensitivity to wind-turbine noise: A basic study in black-tailed gulls. Environmental Engineering Symposium.	Wrong publication type
544368501	Japanese-language record (2021). Mechanisms of vocal learning in songbirds: Insights from noise-avoidance experiments. Japanese Journal of Animal Psychology.	Wrong publication type
544368532	Japanese-language record (2012). Effects of offshore wind power generation on marine ecosystems. Japanese Journal of Conservation Ecology.	Wrong publication type
544326520	Bedanova, I and Chloupek, J and Chloupek, P and Knotkova, Z and Voslarova, E and Pistekova, V and Vecerek, V (2010). Responses of peripheral blood leukocytes to chronic intermittent noise exposure in broilers. BERLINER UND MUNCHENER TIERARZTLICHE WOCHENSCHRIFT. https://doi.org/10.2376/0005-9366-123-186	Wrong language
544326735	Liu, G. and Liu, F.-B. and Lu, S.-W. (2018). The quantitative relationship between road traffic noise and retreat rate of thrush. Chinese Journal of Ecology. https://doi.org/10.13292/j.1000-4890.201812.028	Wrong language
596897786	Nikola Žilinčíková (2016). The impact of noise and light pollution on voice activity of Chiffchaff (Phylloscopus collybita).	Wrong language

Appendix B. Full texts unavailable (n = 3)

These records are retrieval outcomes and are not included among the 170 full-text exclusions.

Rayyan ID	Reference	Primary reason
544368852	Lluch, MDP and Cruz, MCMC (n.d.). Tesis de Diploma. fototeca.uh.cu.	Full text unavailable
544368855	Pablos, MTP (2010). Efectos genéticos y ambientales sobre diferentes indicadores de bienestar en gallinas.	Full text unavailable
544368712	Яковлева, НС and Ермакова, ЛП and Ноздрин, ГА and ... (2022). Способ снижения уровня стресса у перепелов-несушек японского перепела.	Full text unavailable

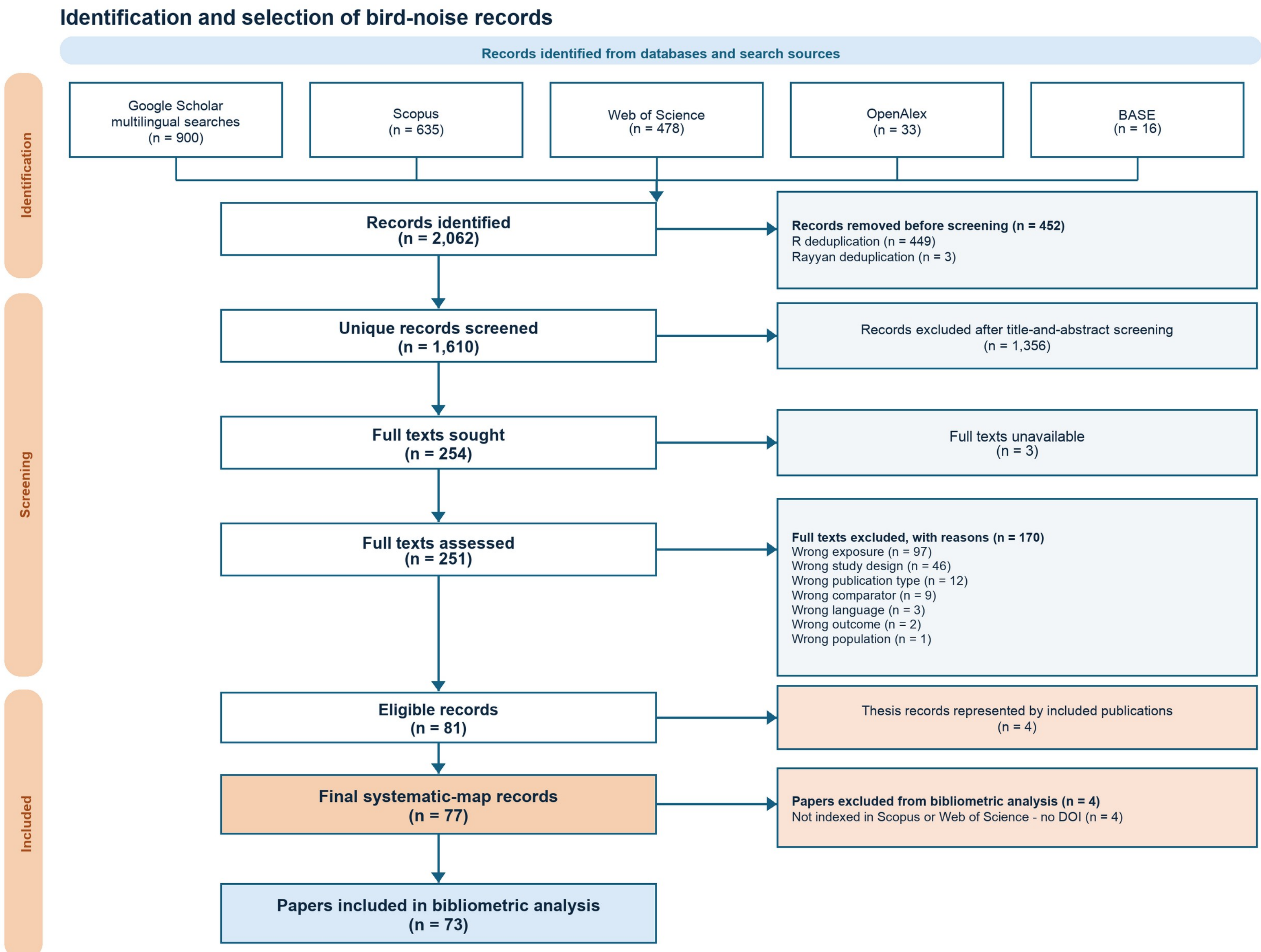

Figure 1. PRISMA-style flow diagram for identification, screening, eligibility assessment, final map reconciliation, and bibliometric inclusion.